



HYDROGRAPH

FGA-2 Fractal Graphene Aggregate Technical Datasheet

HydroGraph 2101 East St Elmo Rd, Ste 150 Austin, TX 78744 +1- (785) 380-4205	
Product Number: FGA-2	Product: Fractal Graphene Aggregate
Product Description	
A few layer, non-functionalized, turbostratic graphene in a powdered form of aggregated nanoplatelets from carbon-rich gas explosion synthesis	

Product Information	
Production Method	Explosion Synthesis
Raw Material	Carbon Bearing Gas
Forms of Materials	PWD – Dry Powder

Characteristic	Test Method	Value
SP2 Bonded Carbon	RAMAN, XPS	Yes (G peak), 80% sp ² (D parameter)
Structural Defects	RAMAN	D/G = 0.3 G width = 50 cm ⁻¹
Number of Layers	RAMAN, X-Ray Diffraction	32-layer average
Z-Axis Dimensions	RAMAN, X-Ray Diffraction	~ 10 nm
Primary Particle Shape	TEM, Light Scattering	Platelets (aggregated)
Lateral Dimensions	TEM	75-200 nm
Aspect Ratio	TEM	1:7.5 to 1:20
Bulk Density	Manual Tapping	70-100 mg/mL*
Chemical/Elemental Analysis	Chemical Analysis	C 98.6%
Oxygen Content %	Chemical Analysis	1.35%
Impurities %	Chemical Analysis	0.1%
Functionalization	Chemical Analysis	Not Detected
Surface Particle Charge	Zeta Potential	33 mV
Graphene Orientation	RAMAN, XRD	Turbostratic
Specific Surface Area (SSA)	BET	25 m ² /g
Crystallinity	Electron Diffraction, X-Ray Diffraction	Crystalline

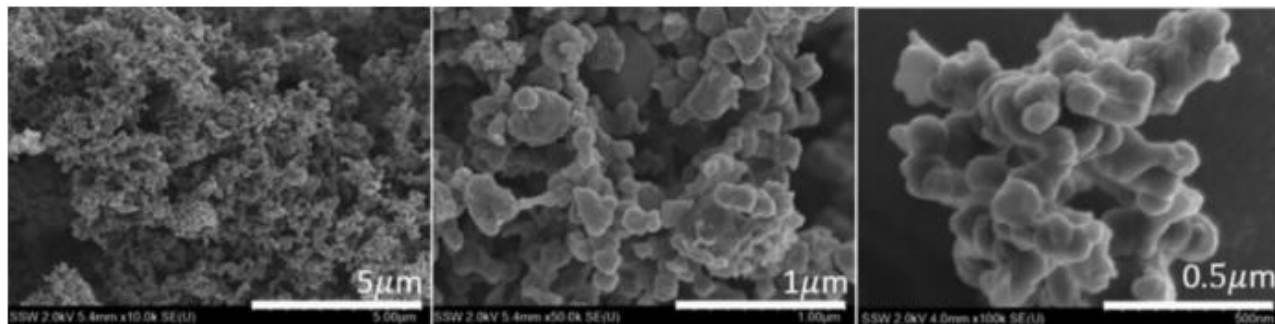
*Density may vary



HYDROGRAPH

Parameters	
Appearance	Black fluffy powder
Number of Layers	~ 32 layers
Lateral Size	Nano-platelets 75 to 200nm.
Shape and Form	Fractal aggregate of nano-platelets
Elemental Analysis	Atomic %: 98.6% Carbon, 1.35% Oxygen, 0.05% Hydrogen, No PAHs
Dispersants/Surfactants	None
Concentration	100%
Solid Content	100%
Solvent content	N/A
Substrate Material	N/A
Sheet Resistance	Not applicable
Color	Black
Odor	None
Solubility in Water	Hydrophobic
Electrical Conductivity	Function of powder compression.

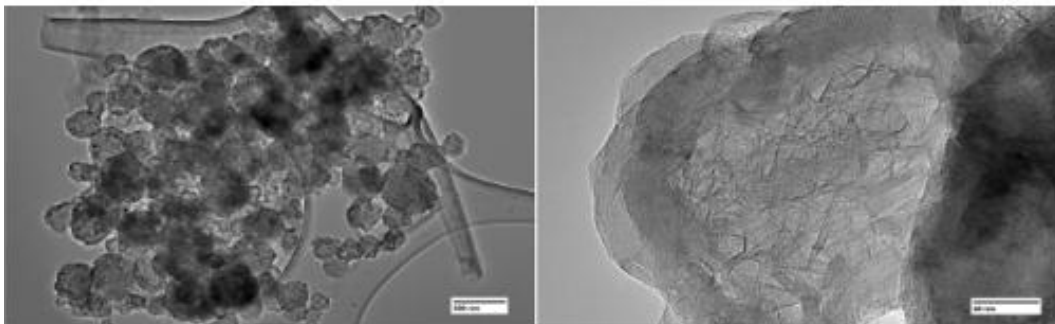
Scanning Electron Micrographs



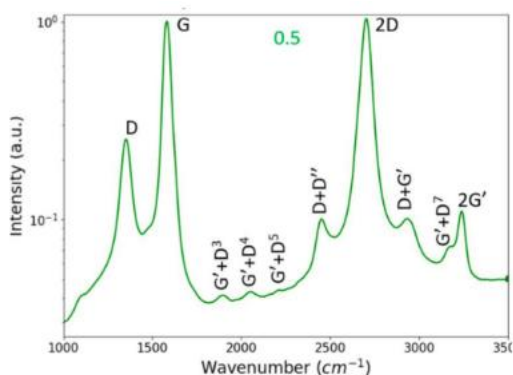


HYDROGRAPH

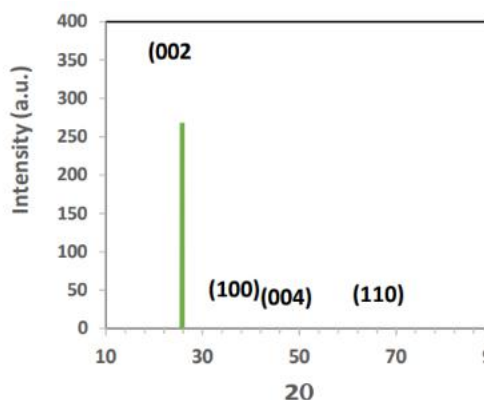
Transmission Electron Micrographs



Raman Spectrum (514 nm)



X-Ray Diffractogram



Notes on Analysis

- The turbostratic nature is indicated by peaks at position 1885cm⁻¹ and 2035cm⁻¹ in the Raman spectrum and the asymmetric (100) peak in the X-Ray Diffractogram.
- AFM- The aggregate nature of our fractal graphene is not amenable to AFM analysis.
- Raman- The nanoscale lateral dimensions of our monomer platelets leads to a high fraction of defect edge sites which enhance the intensity of the Raman D bands.

Disclaimer: The technical information and other statements contained in this document are based upon tests Hydrograph believes are reliable, but the accuracy or completeness of such information is not guaranteed. Given the variety of factors that can affect the use and performance of this product, user is solely responsible for evaluating the Hydrograph product and determining whether it is fit for a particular purpose and suitable for user's method of application.